

Whom Are We Really Predicting? Population Definition Inflates Early-Warning Performance on OULAD—A Time-Aware, Explainable Re-Examination

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Abstract—Early-warning models for at-risk students are routinely evaluated on every enrolment in a course—including students who have already withdrawn by the time the prediction is made. We show on the Open University Learning Analytics Dataset (OULAD) that this population choice materially inflates early-warning claims. Using a leakage-safe pipeline with a frozen, group-aware student split, cumulative features at six course-progress checkpoints, and five benchmarked algorithms, we re-score every headline result on two cohorts: all 32,593 enrolments (the frame used by prior work) and the students still enrolled at each checkpoint (the only frame an intervention can reach). Three findings emerge. First, at 40% of course length the full-cohort recall of the best model, XGBoost, is 0.811 (95% CI [0.798, 0.824]), but still-enrolled recall is only 0.678 (95% CI [0.656, 0.698]); the cohort gap 0.133 (95% CI [0.122, 0.144]) excludes zero at every checkpoint (1,000-resample bootstrap). The reliability criterion (recall ≥ 0.80 and PR-AUC ≥ 0.80) is met on the still-enrolled cohort only at course end (0.841 (95% CI [0.821, 0.861])). Second, gradient-boosted trees dominate the five-algorithm benchmark, with differences confirmed by Friedman and Wilcoxon tests over 25 cross-validation fits. Third, SHAP explanations are stable across seeds (top-10 Jaccard 0.69), far above a Monte-Carlo null baseline (mean 0.119, 99th percentile 0.333). We propose dual-cohort reporting as a minimal standard for early-warning research.

Index Terms—learning analytics, at-risk prediction, OULAD, explainable AI, SHAP, LIME, evaluation methodology, population definition

I. Introduction

Virtual learning environments (VLEs) now mediate a large share of higher-education teaching, and every action a student takes—opening a resource, submitting an assessment, visiting a forum—leaves a digital trace. Failure and drop-out nevertheless remain persistent problems in online and distance education, and learning analytics has responded with a rich body of work on at-risk prediction, from systematic comparisons of supervised algorithms [1] to prediction at multiple points of course length [2].

Three limitations recur in this literature. The first is opacity: the strongest models behave as black boxes, so instructors cannot see why a student is flagged; systematic reviews of explainable student-performance prediction

identify this as a central obstacle to adoption [3]. The second is lateness: many studies predict at or near the end of the course, when intervention is no longer possible [2]. The third is class imbalance, to which we add a factual correction: under the standard OULAD label mapping (at-risk = {Fail, Withdrawn} versus {Pass, Distinction}), the at-risk class is not a rare minority but a slight majority—52.8% of enrolments, an imbalance ratio of 1.12—so resampling addresses a problem this dataset does not actually have.

This paper argues that a fourth limitation, so far unexamined, matters more than any of these: the definition of the evaluation population. Time-aware studies fix each enrolment’s label from its end-of-course outcome and score models at intermediate checkpoints on the full enrolment roster. At any mid-course checkpoint that roster contains students who have already withdrawn; their near-total inactivity makes them trivially recognisable, so a portion of the measured “early-warning” performance consists of detecting outcomes that have already happened rather than forecasting ones that have not. This is not label leakage—the label never enters the features—but an estimand mismatch: end-of-course outcome classification is well defined on all enrolments, whereas early warning for intervention is meaningful only on students a tutor can still reach. No prior OULAD study we reviewed separates the two populations, and the size of the resulting inflation has never been quantified.

We quantify it. On OULAD [4] we build a leakage-safe, time-aware pipeline—cumulative features at six checkpoints (10–100% of course length), five algorithms from three model families, a frozen group-aware student-level split—and re-score every headline result on both cohorts, attaching nonparametric confidence intervals from 1,000 bootstrap resamples. Because explanations are part of an early-warning product, we also evaluate SHAP [6] and LIME [7] explanations quantitatively, calibrating every agreement statistic against an explicit Monte-Carlo null baseline. OULAD is chosen deliberately: as the field’s standard public benchmark it makes the population-

definition effect directly relevant to the largest body of comparable results, and its fixed anonymised snapshot makes every number in this paper reproducible from committed code and data.

Concretely, the study answers three questions, each stated per cohort where the distinction matters. First, which of five candidate algorithms gives the best at-risk prediction at each checkpoint, and how early can a prediction be called reliable under an explicit criterion—recall ≥ 0.80 and PR-AUC ≥ 0.80 on the at-risk class of a sealed test set? Second, how consistent are the explanations produced by SHAP and LIME for the same model, and how does their stability behave across random seeds, across the six checkpoints, and across training regimes? Third, what does imbalance handling (class weighting, SMOTE, ADASYN) do to both predictive accuracy and the explanations a tutor would be shown, given that the at-risk class is in fact a slight majority?

The contributions are fourfold:

- Quantified population-definition inflation. A dual-cohort re-scoring of the same predictions with student-level bootstrap confidence intervals shows a recall gap between the full and still-enrolled cohorts whose 95% interval excludes zero at every checkpoint; the reliability criterion moves from 40% of course length (full cohort, borderline) to course end (still-enrolled).
- A verified time-aware benchmark. Five algorithms across three families are compared under an identical leakage-safe protocol at six checkpoints, with the ranking validated by repeated cross-validation (25 fits) and Friedman/Wilcoxon–Holm testing—closing the ensemble gap left by prior comparisons [1].
- Explanation stability with a null baseline. A quantitative stability framework (top- k Jaccard with $k = 10$, Spearman rank correlation, across seeds, explainers, checkpoints, and training regimes) whose agreement statistics are calibrated against a Monte-Carlo null distribution—addressing the qualitative-only assessment gap in educational XAI [3], [8].
- A dual-reporting rule and an operational tool. A reporting discipline—every earliness claim names its cohort, every explanation carries its checkpoint—implemented end-to-end in an instructor dashboard whose worklist is restricted to still-enrolled students.

Section II reviews related work and contrasts evaluation protocols. Section III describes the data and method. Section IV reports the benchmark and the dual-cohort results. Section V analyses explanation stability. Sections VI and VII discuss limitations and conclude.

II. Related Work

At-risk prediction on OULAD. OULAD [4] is a public, anonymised benchmark of 32,593 student registrations on 22 module-presentations of The Open University, comprising demographics, a VLE clickstream, and assessment

records in seven relational tables; its open licence has made it the standard testbed for reproducible at-risk prediction. Tomasevic et al. [1] systematically compared supervised techniques for exam-performance prediction on a subset of the DDD module (3,166 students after excluding non-exam-takers), finding that an ANN with combined engagement and performance features performed best and that demographics added little once behavioural data were available—a result that informs our feature prioritisation. Methodologically, rows with missing values were dropped (silently excluding exactly the non-submitting students an early-warning system most needs to see), evaluation used random 80:20 or 60:20:20 splits at the row level, and gradient-boosted ensembles were absent from the comparison. Adnan et al. [2] provide the template for time-aware prediction: cumulative demographic, clickstream, and assessment features on the full OULAD at 20–100% of course length, evaluated with 10-fold cross-validation, with predictions becoming acceptably reliable from roughly 40–60% of course length—the empirical basis of our checkpoint schedule, which adds a 10% point. For imbalance they merged outcome classes (Pass+Distinction versus Fail+Withdrawn) rather than resampling, and no group-aware split keyed on the student identifier was applied in either study.

Explainable AI in education. SHAP [6] unifies additive feature-attribution methods on a Shapley-value foundation; LIME [7] fits an interpretable surrogate in the neighbourhood of a single prediction. Reviewing explainable student-performance prediction, Alamri and Alharbi [3] found SHAP and LIME dominant but observed that explanations are almost always assessed qualitatively, by visual inspection of importance plots. Gunasekara and Saarela [8] applied both explainers to an ANN and a decision tree on a three-module OULAD subset (17,091 samples, 5-fold cross-validation repeated 50 times); their assessment of the local explanations likewise remains mainly qualitative. A quantitative measure of explanation stability—whether an explainer reproduces the same ranking across seeds, time points, or training regimes—is largely absent, and no study we reviewed supplies a chance baseline against which an agreement score could be judged.

Imbalance handling. SMOTE [9] and its adaptive variants such as ADASYN synthesise minority-class examples by interpolating between neighbouring minority instances; class weighting offers a resampling-free alternative by rescaling the loss. The OULAD literature treats imbalance inconsistently—class-merging in [2], exclusion of non-completers in [1]—while the studies in our reviewed corpus that tackle imbalance head-on with resampling do not use OULAD at all. No study we reviewed measures what resampling does to explanations, the interaction our third question targets.

Protocol comparison. Following a concept-centric review [5] of 27 papers (2019–2026), we cross-tabulated prior work along three axes—time-awareness, explainability,

imbalance handling—and found no study covering all three on OULAD: the time-aware studies [1], [2] provide no explanations, the XAI study [8] considers no multiple time points, and the resampling-focused work in the corpus uses other datasets. More consequential for the present argument are the evaluation protocols, contrasted in Table I: the base studies split at the row level, so a student appearing in several module-presentations can occupy both train and test partitions; none separates withdrawn students from the evaluation population at intermediate time points; and none attaches uncertainty estimates to its headline claims. These three choices are precisely the ones this paper revisits.

III. Data and Method

A. Dataset

OULAD [4] is distributed as a fixed, anonymised snapshot under a CC-BY 4.0 licence, guaranteeing byte-identical inputs across researchers. Its seven relational tables cover 32,593 enrolments (28,785 distinct students) on 22 module-presentations; the unit of analysis is one enrolment, keyed by (student, module, presentation), and post-join verification confirms that all joins preserve exactly 32,593 rows with no duplicate keys. The behavioural core is a clickstream of approximately 10.6 million rows recording daily interaction counts with VLE materials; its per-day date column, together with the submission dates in the assessment table, is the architectural foundation of the time-aware design, since every time-dependent feature can be recomputed at an arbitrary cutoff. One documented source-data quirk: 7.4% of clickstream rows are exact duplicates with no distinguishing key, so the schema cannot separate a double-logging artefact from a legitimately repeated record. Because deleting one copy of an indistinguishable pair would be an unverifiable guess, and the baseline studies work from the unduplicated tables, the pipeline retains them and documents possible over-counting as a limitation (Section VI).

B. Target and the Two Estimands

The binary target maps the final outcome to at-risk = 1 for {Fail, Withdrawn} and 0 for {Pass, Distinction}; Distinction is merged into Pass because the task is risk detection, not achievement ranking. Under this mapping the at-risk class covers 52.8% of enrolments (imbalance ratio 1.12)—a slight majority, so imbalance handling is studied as a controlled robustness question rather than the rescue of a rare class.

For prediction over time we adopt a fixed label, fixed population design: every enrolment keeps its end-of-course label at every checkpoint, and the student roster is identical across checkpoints, so performance curves compare identical populations that differ only in available information. A student who withdrew before checkpoint t remains in the t dataset, labelled at-risk; the temporal cut removes all post-withdrawal events, so the observed

decline in engagement is genuine early-warning signal, not an error.

This fixed label supports two distinct estimands. End-of-course outcome classification—“will this enrolment end in Fail or Withdrawn?”—is well defined on all 32,593 enrolments at every checkpoint and is the frame in which prior work operates. Early warning for intervention—“whom should a tutor contact at checkpoint t ?”—is meaningful only on the subpopulation still enrolled at the cutoff, because an intervention cannot reach a student who has already left. On the full cohort, part of any model’s measured strength is recognition of already-withdrawn students rather than forecasting; this motivates the paper’s reporting rule that every reliability claim names its cohort, and its headline analysis re-scores every result on both frames.

C. Features, Checkpoints, and Preprocessing

Twenty-eight raw features are engineered per enrolment in three groups: demographics and context (11 static features, including registration timing and module identity), engagement (13 clickstream aggregates such as total clicks, active days, eight per-activity-type counts, peak and mean daily intensity, and days since last activity), and performance (4 assessment aggregates—mean and weighted cumulative scores, submission count, and a not_submitted indicator that keeps imputed zero scores distinct from genuine zero-score submissions). All time-dependent features are cutoff-driven: at checkpoint t , only clickstream and submission records dated on or before the checkpoint day enter the aggregation, so each feature answers “what was known about this student at t ”. Because course lengths differ across the 22 module-presentations, checkpoints are defined in relative course progress— $t \in \{10, 20, 40, 60, 80, 100\}\%$ of each presentation’s length, extending the schedule of [2] with a 10% point—and materialised once in a committed checkpoint map, producing six datasets over the same 32,593 enrolments. One defect in the cutoff logic was found during an internal audit and fixed: assessments banked from a previous presentation were counted as due but not as submitted, mis-setting the submission indicator for 78 of 32,593 enrolments (0.24%) at $t = 100\%$; the fix is pinned by a regression test, and every table in this paper is built from results recomputed under the fixed pipeline.

Preprocessing follows a strict split-first order (split, missing values, outliers, encoding and scaling, resampling), with every learned statistic fitted on the training partition only. Missing assessment aggregates become explicit “no submission yet” signals rather than being dropped; right-skewed click counts are log-transformed and heavy tails winsorised at training-fold percentiles; and a single fitted transformer (standard scaling for numeric, ordinal encoding with fixed orders, one-hot encoding without a dropped reference level so explainers can attribute importance to every category) expands the 28 raw features into a 49-

TABLE I

Evaluation protocols of prior OULAD studies and this work. Caveat: reported performance figures are not directly comparable across rows, because the studies differ in cohort definition, split unit, validation scheme, and label treatment.

Study	Population / cohort	Split unit and validation	Withdrawn completers / non-completers	Uncertainty reported
Tomasevic et al. [1]	DDD-module subset (two presentations; 3,166 students after excluding non-exam-takers)	Random 80:20 (60:20:20 for the ANN), row-level; averages over repeated trials	Rows with missing values dropped; non-submitters excluded from the cohort	None
Adnan et al. [2]	Full OULAD (22 module-presentations; 32,593 enrolments)	10-fold cross-validation, row-level (85/15 split for the deep model)	Merged into the at-risk class (Pass+Distinction vs. Fail+Withdrawn); never evaluated separately	None
Gunasekara and Saarela [8]	Three-module subset (17,091 samples, 14 attributes)	5-fold cross-validation repeated 50 times, row-level	Classes merged; not evaluated separately	None (explanations assessed qualitatively)
This work	Full OULAD (32,593 enrolments; 28,785 students)	Frozen group-aware student-level 80:20 hold-out (zero student overlap) + 5-fold $\times$ 5-seed CV on the training partition	Kept with fixed label (comparable frame) and re-scored on the still-enrolled cohort at every checkpoint	95% bootstrap CIs (1,000 resamples); Friedman and Wilcoxon-Holm tests

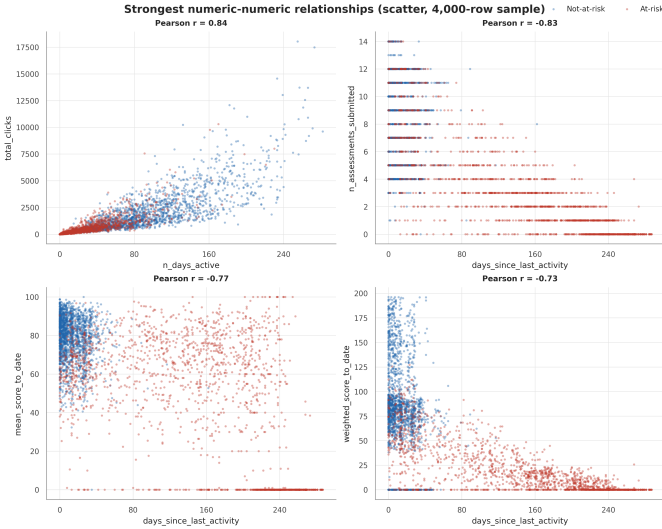


Fig. 1. The four strongest numeric–numeric feature relationships as class-coloured scatter plots (fixed-seed 4,000-row sample). The strongest couplings are volume effects (e.g., active days vs. total clicks); no feature approaches the $|r| \geq 0.95$ leakage-alarm threshold against the target.

column model matrix. Exploratory analysis confirmed that engagement and performance dominate the class signal while demographics associate only weakly with the target, that no feature approaches the $|r| \geq 0.95$ leakage-alarm threshold against the label, and that the strongest inter-feature relationships are expected volume couplings.

D. Leakage Prevention

Four rules keep every checkpoint dataset honest about prediction-time information: (1) an assessment submission is retained only if submitted on or before the cutoff day; (2) a clickstream row is retained only if dated on or before

the cutoff; (3) early-withdrawn students are retained with their fixed label—the label is derived solely from the final outcome and never enters the features, so this violates no information constraint (its estimand consequences are handled by dual-cohort scoring); and (4) every data-dependent component—imputers, outlier thresholds, encoders, scaler, resampler—is fitted on the training fold only. These rules are enforced by 21 automated tests that assert, among other properties, that no retained record postdates its cutoff, that the split has zero student overlap, and that a feature allow-list blocks leaky columns from the model matrix; all tests must pass before any modelling step runs.

E. Split, Validation, and Model Selection Protocol

Because a student can appear in several module-presentations, a row-level split leaks students across the partition. We therefore use a group-aware, stratified 20% hold-out split by student identifier, derived once and frozen: the canonical test-student list is committed to the repository, and the split harness only ever loads this list—re-derivation under a different library version would silently change the partition and invalidate every published number. The verified properties hold identically at all six checkpoints: 26,104 training rows and 6,489 test rows from 5,756 test students, zero student overlap, and a preserved class ratio. Model selection uses 5-fold cross-validation repeated over 5 seeds (25 fits per model) on the training partition, group-aware and stratified, with preprocessing re-fitted inside every fold; the global seed is 42.

Five algorithms spanning three families are benchmarked with near-default configurations: logistic regression (linear reference), random forest (bagging), XGBoost and LightGBM (gradient boosting), and a two-hidden-

layer multilayer perceptron with early stopping (non-tree, non-linear). Covering three families ensures the ranking is not an artefact of one inductive bias, and all five appear in the OULAD base studies, keeping results comparable. No per-model tuning is performed at the selection stage, so no candidate is favoured by attention; a subsequent randomised search (40 configurations, group-aware CV) confirmed that tuning buys only marginal PR-AUC and no recall improvement, so near-default settings are retained throughout—consistent with the observation that when five differently biased learners sit within a few points of one another, the feature signal, not the estimator configuration, is the binding constraint.

The recall-first evaluation is a cost argument, not a convention: a missed at-risk student forfeits the chance to intervene, whereas a false alarm costs a tutor conversation. Accuracy is deliberately demoted because, with the at-risk class a slight majority, it flatters trivial classifiers; PR-AUC complements recall as the threshold-free summary of the precision–recall trade-off on the positive class. A checkpoint is deemed reliable when the model reaches both recall ≥ 0.80 and PR-AUC ≥ 0.80 on the at-risk class of the held-out test set.

F. Uncertainty Quantification and Null Baselines

Two inferential devices distinguish this study from the base protocols. First, every headline test-set metric carries a nonparametric 95% confidence interval from the percentile bootstrap with 1,000 resamples of the frozen test set, resampled at the student level so that all enrolments of a drawn student enter together and the grouped structure is respected. The same resamples are scored on both cohorts, and the per-resample difference yields an interval for the between-cohort recall gap that accounts for the correlation between the two estimates. Model-ranking claims are tested separately with a Friedman test over the 25 paired cross-validation fits, followed by pairwise Wilcoxon signed-rank tests with Holm correction.

Second, explanation-agreement statistics are calibrated against an explicit chance baseline. A top-10 Jaccard overlap of, say, 0.4 is uninterpretable in isolation: even unrelated rankings share features by chance. We estimate the null distribution by Monte Carlo—repeatedly drawing two independent uniformly random top-10 subsets of the 49-feature space and recording their Jaccard overlap—obtaining a null mean of 0.119 with 95th and 99th percentiles of 0.25 and 0.333. An observed agreement is claimed as genuine stability only if it exceeds the 99th percentile of this null.

IV. Results

A. Benchmark at Course End

Every model passes through an identical four-step protocol: load the frozen split, fit all preprocessing on the training partition only, fit the model, and score the

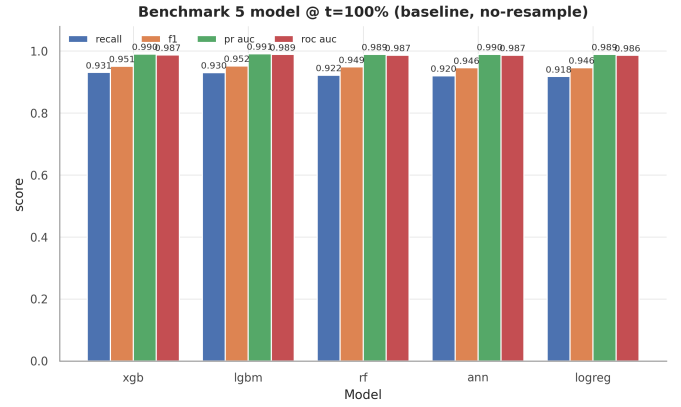


Fig. 2. Baseline benchmark of the five candidate models at $t = 100\%$ on the held-out test set: recall, F1, PR-AUC, and ROC-AUC per model (no resampling, shared seed 42).

untouched test set on seven metrics—recall on the at-risk class, precision, F1, PR-AUC, ROC-AUC, balanced accuracy, and the Brier score for probability quality. Table II reports the resulting comparison of the five models at $t = 100\%$, without resampling; these baseline rows are also the reference point against which the imbalance-handling regimes of Section IV-F are compared, so the before–after contrast is well defined.

XGBoost tops the recall-first ranking with an at-risk recall of 0.930 and a PR-AUC of 0.990 (0.990 (95% CI [0.989, 0.992])). Two structural observations matter more than the winner’s identity. First, the three tree ensembles track each other closely on all seven metrics—the boosting family, not one implementation, dominates. Second, logistic regression trails by only a few points, indicating that the engineered features carry a strong, largely linearly separable signal on which the ensembles add a real but modest increment.

B. Is the Ranking Real?

Small between-model gaps demand verification. Table III summarises the repeated cross-validation (5 folds $\times$ 5 seeds, 25 fits per model, preprocessing re-fitted per fold) as mean $\pm$ standard deviation.

XGBoost leads cross-validated recall at 0.9309 ± 0.0047 , with LightGBM immediately behind, and the CV ranking reproduces the held-out ranking—the single-split benchmark is not an outlier. The Friedman test over the 25 paired fits rejects equal performance far below the 0.05 level on every metric, so the small observed differences are systematic rather than noise; Holm-corrected pairwise Wilcoxon signed-rank tests sharpen the picture, showing XGBoost significantly ahead of each of the other four models on recall while LightGBM leads the composite metrics (F1, PR-AUC, ROC-AUC). Given the recall-first objective and the exact TreeExplainer attributions required by the explanation analysis, XGBoost is selected as the primary model.

TABLE II

Baseline benchmark of the five candidate models at $t = 100\%$ on the frozen held-out test set (no resampling, shared seed 42), ranked recall-first on the at-risk class.

model	roc_auc	pr_auc	f1	recall	precision	balanced_acc	brier
ann	0.9868	0.9897	0.9477	0.9259	0.9705	0.9477	0.0404
lgbm	0.9889	0.9913	0.9511	0.9295	0.9736	0.9511	0.0368
logreg	0.9857	0.9890	0.9464	0.9171	0.9776	0.9471	0.0414
rf	0.9868	0.9896	0.9466	0.9165	0.9788	0.9475	0.0405
xgb	0.9872	0.9902	0.9503	0.9298	0.9718	0.9503	0.0391

TABLE III

Repeated cross-validation on the training partition (5-fold $\times$ 5 seeds; 25 fits per model, $t = 100\%$): mean $\pm$ standard deviation per model and metric.

model	recall_mean	recall_std	f1_mean	f1_std	pr_auc_mean	pr_auc_std
lgbm	0.9295	0.0060	0.9516	0.0036	0.9907	0.0008
xgb	0.9309	0.0047	0.9507	0.0032	0.9901	0.0008
rf	0.9202	0.0051	0.9489	0.0030	0.9889	0.0009
ann	0.9245	0.0081	0.9463	0.0038	0.9891	0.0009
logreg	0.9180	0.0054	0.9452	0.0033	0.9887	0.0010

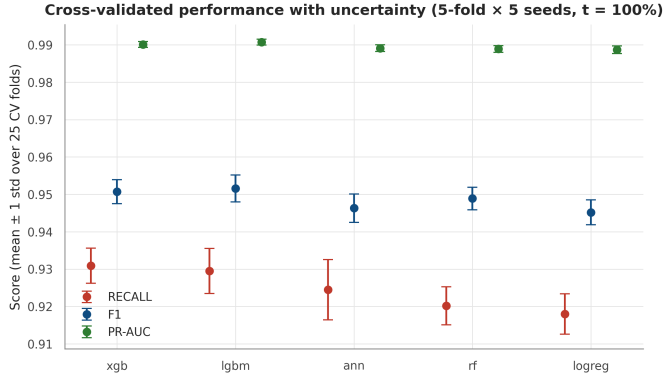


Fig. 3. Cross-validated recall, F1, and PR-AUC per model as mean $\pm$ one standard deviation over the 25 fits (5-fold $\times$ 5 seeds, $t = 100\%$): between-model gaps exceed within-model variability.

In bias-variance terms the two verification layers also explain why the ranking is trustworthy. The per-model standard deviations are small relative to between-model gaps—the estimators are low-variance at this sample size—and overfitting is bounded by construction: every score is computed on data the model never saw (25 held-out folds plus the frozen test set), preprocessing is re-fitted inside each fold, and the close agreement between cross-validated and held-out rankings confirms that no model is merely memorising its training fold. Underfitting is visible instead as the systematic few-point gap between the linear baseline and the tree ensembles: logistic regression’s higher-bias hypothesis class cannot represent the feature

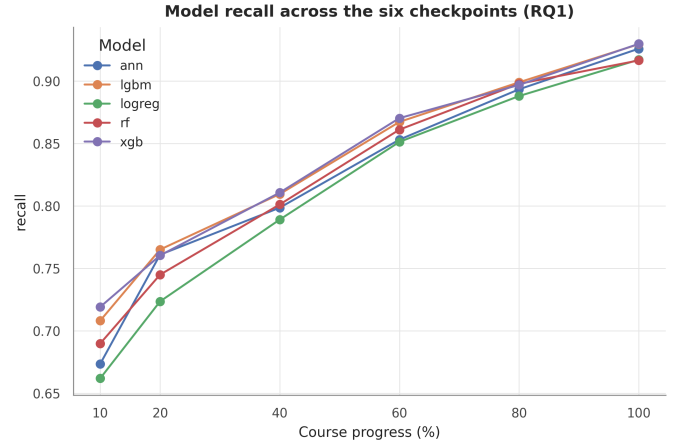


Fig. 4. At-risk recall of the best model per checkpoint across the six course-progress checkpoints on the full test cohort; the horizontal line marks the recall ≥ 0.80 reliability criterion.

interactions the boosted trees exploit, and that difference is precisely the increment the ensembles add.

C. Time-Aware Performance on the Full Cohort

The identical protocol was repeated at all six checkpoints on the same frozen test students.

LightGBM is narrowly ahead at the two earliest checkpoints (both below the reliability bar) and XGBoost from $t = 40\%$ onward. On the full enrolment cohort, the recall point estimate crosses 0.80 at $t = 40\%$, reaching 0.811, and rises monotonically to 0.930 at course end,

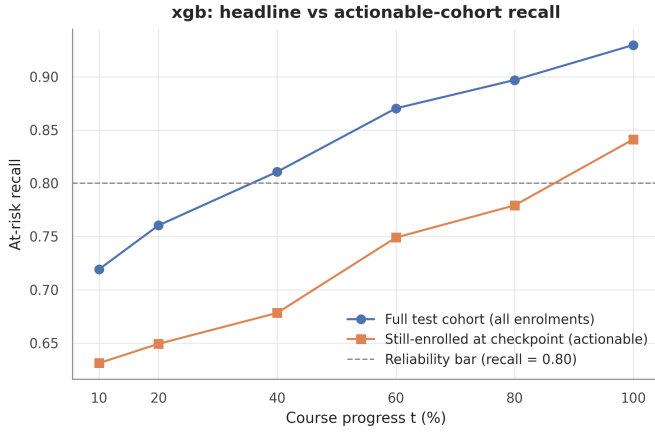


Fig. 5. XGBoost at-risk recall per checkpoint on the full test cohort versus the still-enrolled subpopulation, with the recall ≥ 0.80 criterion marked; the gap between the curves quantifies the contribution of already-withdrawn students to measured performance.

with PR-AUC comfortably above its criterion over the same range—so recall, not ranking quality, is the binding constraint on earliness. The trajectory is consistent with the 40–60% reliability window of [2] on the same dataset. The bootstrap, however, imposes a nuance invisible in point-estimate reporting: the interval at $t = 40\%$ is 0.811 (95% CI [0.798, 0.824]), whose lower bound touches the criterion. The honest full-cohort statement is therefore that reliability at 40% of course length is plausible but borderline under sampling uncertainty, firming up only at later checkpoints (at course end, 0.930 (95% CI [0.921, 0.939])).

D. Whom Are We Really Predicting? The Dual-Cohort Result

The full cohort is only one of the two estimands of Section III-B. Because the at-risk label includes withdrawal and the roster is fixed, the test set at every checkpoint contains students who have already left—923 test enrolments before the earliest checkpoint at $t = 10\%$, and more at each later cutoff. For these records the model records an outcome that has already happened. Table IV therefore re-scores the same XGBoost predictions on the still-enrolled subpopulation at each checkpoint, with bootstrap intervals for both cohorts and for their gap.

The gap is substantial, systematic, and statistically unambiguous. On the still-enrolled cohort, recall is 0.678 at $t = 40\%$ (0.678 (95% CI [0.656, 0.698])) and 0.779 at $t = 80\%$, and reaches the criterion only at course end: 0.841 (0.841 (95% CI [0.821, 0.861])). The bootstrap interval for the between-cohort gap excludes zero at every checkpoint—0.133 (95% CI [0.122, 0.144]) at $t = 40\%$ and still 0.089 (95% CI [0.076, 0.101]) at $t = 100\%$, where the residual gap reflects that at-risk students still enrolled at course end (imminent failers rather than

withdrawers) are intrinsically harder to detect. LightGBM reproduces the same qualitative shape, so the finding is not specific to one model. Note that nothing about the model changes between the two rows of any checkpoint pair: the predictions are identical and only the evaluation population changes. The difference between the curves is a pure population effect—exactly the quantity a reader needs in order to translate a published full-cohort number into an expectation about intervention yield.

The dual-cohort answer to the earliness question is therefore stated in two parts, always together. On the full enrolment cohort—the frame in which the base studies [1], [2] operate—XGBoost is the best model from $t = 40\%$ onward and reliability is plausible, though borderline under the bootstrap, from that checkpoint. On the still-enrolled cohort—the frame in which a tutor can still act—the same criterion is met only at course end, and mid-course predictions, while informative, fall short of the bar.

Two clarifications guard against misreading. First, this is not data leakage: the label never enters the features, and the near-total inactivity of an early-withdrawn student is genuinely observed behaviour. Second, it is a population-definition result: part of full-cohort performance at every checkpoint comes from recognising students who have already gone—legitimate for outcome classification, but not early warning. The answer to which checkpoint yields “reliable” prediction thus changes by three checkpoints depending on a population choice that prior work does not report, which is precisely why we propose that both frames always be published together.

E. Operating Thresholds and Subgroup Fairness

A deployed system needs a decision threshold, and choosing it on the test set would optimistically bias every reported metric. Thresholds are therefore selected on out-of-fold validation predictions from cross-validation on the training partition, frozen, and scored on the test set exactly once per policy. The F1-optimal policy selects 0.56, nearly coinciding with the default 0.5, and its validation metrics transfer to the test set with negligible drift—retrospective evidence that the default-threshold results above were not tuned on the test set. An institutional high-recall policy is also quantified: mandating recall ≥ 0.90 yields a validation-chosen threshold of 0.86 that delivers the mandated recall on the test set at a precision of 0.993—an operating point at which flagged students are almost always genuinely at risk. The full validation trade-off curves are committed alongside the result tables, so an institution can select any other operating point without ever touching the test set.

For the selected model at $t = 100\%$ and the default threshold, recall and false-positive rate were disaggregated over every level of six demographic attributes (gender, region, deprivation band, prior education, age band, declared disability; subgroups with at least 50 test members). Table V summarises the within-attribute ranges: the

TABLE IV

Dual-cohort re-scoring of the selected XGBoost model per checkpoint: at-risk recall on the full test cohort and on the still-enrolled subpopulation, with 95% percentile-bootstrap confidence intervals (1,000 student-level resamples of the frozen test set) and the between-cohort recall gap.

t (%)	Full cohort	Still-enrolled	Gap
10	0.719 (95		
20	0.760 (95		
40	0.811 (95		
60	0.870 (95		
80	0.897 (95		
100	0.930 (95		

TABLE V

Within-attribute subgroup gaps at $t = 100\%$ and the default threshold: range of at-risk recall and false-positive rate across the levels of six demographic attributes (subgroups with ≥ 50 test members).

attribute	n_levels	recall_min	recall_max	recall_gap	fpr_gap
imd_band	11	0.8941	0.9599	0.0658	0.0464
region	13	0.9091	0.9502	0.0411	0.0385
highest_education	5	0.9264	0.9565	0.0301	0.0526
gender	2	0.9147	0.9409	0.0262	0.0021
disability	2	0.9279	0.9446	0.0167	0.0210
age_band	2	0.9183	0.9342	0.0159	0.0070

largest recall gap across all attributes is 6.6 percentage points, observed on imd_band, with false-positive-rate gaps of similarly modest order and no subgroup—including students declaring a disability—falling dramatically below the overall recall level. These are observed disparities at one checkpoint and threshold, not a fairness certification; a deployment should monitor the same disaggregated metrics continuously.

F. Robustness to Imbalance Handling

Because the at-risk class is a slight majority (52.8%, ratio 1.12), resampling here answers a robustness question, not a rare-class rescue—indeed, default-configured SMOTE [9] and ADASYN oversample the not-at-risk class on this dataset, the opposite of the textbook “rescue the rare at-risk class” intuition, a behaviour we detected early and document rather than let pass unremarked. Every resampling or weighting intervention is applied to the training partition only, after preprocessing has been fitted; the held-out test set is never resampled and

always reflects the real class distribution. Retraining all five models at $t = 100\%$ under four regimes (none, class weighting, SMOTE, ADASYN) moves no primary metric materially: the largest spread on at-risk recall, F1, or PR-AUC across all models and strategies is 0.0069, of the same order as seed-to-seed cross-validation variation, with no strategy consistently ahead. This is the expected behaviour at an imbalance ratio of 1.12: with both classes abundantly represented, neither synthetic oversampling nor loss reweighting has room to move the decision boundary. Predictive conclusions on OULAD under this label mapping therefore do not depend on the imbalance-handling choice; Section V-C shows the same holds for explanations.

V. Explanation Stability

A. What Drives an At-Risk Flag

Global SHAP values for the selected model are computed with TreeExplainer, which is exact for tree ensembles [6].

One feature dominates by a wide margin: days_since_last_activity, the length of the student’s current silence in the VLE, followed by weighted_score_to_date, the cumulative weighted assessment score. Attribution directions match tutor intuition—long silence pushes strongly toward a flag, a high running score pulls away—and the remaining contributors (submission counts, module identity, per-activity click features) act in similarly plausible directions, so the model’s strongest signals are precisely the two indicators human tutors already watch: disengagement

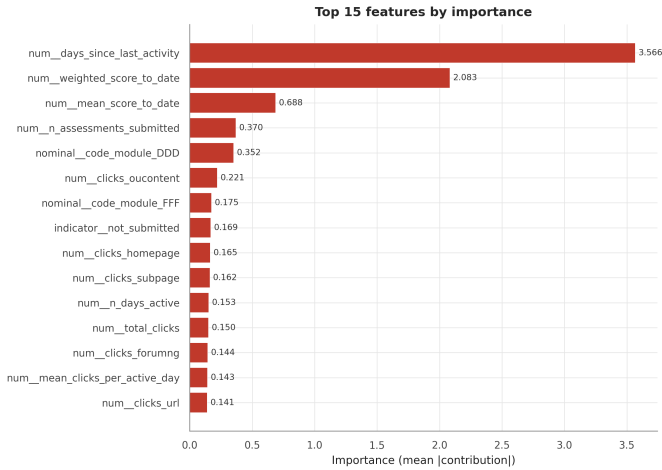


Fig. 6. Global SHAP feature importance (mean absolute SHAP value) for XGBoost at $t = 100\%$. The length of the student’s current VLE silence dominates, followed by the cumulative weighted assessment score.

and weak assessment performance. The dominance of a silence feature is also an inside-the-model corroboration of Section IV-D: on the full cohort—where recall is 0.930 versus 0.841 on the still-enrolled subgroup—`days_since_last_activity` is exactly the channel through which already-disengaged and withdrawn students are recognised rather than forecast. The explanation layer thus confirms, mechanistically, why every headline claim must name its cohort. Because global importances do not help a tutor with an individual student, LIME [7] supplies the local “why”—the form in which an explanation is actually consumed in an intervention conversation, and the form served per student by the dashboard.

B. Stability Against the Null Baseline

An explanation can support intervention decisions only if it is reproducible, and an agreement score can support a stability claim only if it is compared against chance. All agreement statistics below are therefore read against the Monte-Carlo null of Section III-F: two random top-10 draws from the 49-feature space overlap with mean Jaccard 0.119, 95th percentile 0.25, and 99th percentile 0.333. Without this calibration, the partial SHAP–LIME overlap reported below could be spun as either reassuring or alarming; against the null, its meaning is unambiguous.

Across seeds. Retraining the selected model under five seeds and re-explaining yields a mean top-10 Jaccard of 0.69 and a mean Spearman rank correlation of 0.97 across seed pairs—far above the null’s 99th percentile. The global feature story is a property of the data and model class, not an artefact of one training run.

Across explainers. SHAP and LIME agree partially: the Jaccard overlap of their global top-10 sets is 0.43, computed on a LIME sample of $n = 100$ test students. This too exceeds the null’s 99th percentile—the agreement

TABLE VI
Pairwise agreement of global SHAP importance rankings for XGBoost at $t = 100\%$ trained under four imbalance-handling regimes (top-10 Jaccard; Spearman over the full 49-feature ranking).

strategy_a	strategy_b	jaccard_top10	spearman
none	class_weight	0.5385	0.9739
none	smote	0.6667	0.9708
none	adasyn	0.8182	0.9674
class_weight	smote	0.6667	0.9709
class_weight	adasyn	0.6667	0.9641
smote	adasyn	0.8182	0.9744

is genuine—but it is far from unity, and the mechanism is methodological rather than pathological: SHAP allocates exact Shapley contributions of the trained model, whereas LIME fits a local linear surrogate to perturbed inputs and perturbs one-hot categories as if continuous, injecting noise into mid-ranking features. The two explainers agree on the headline drivers and diverge in the tail. Our policy: anchor global claims on SHAP (exact for trees), use LIME as a per-student local cross-check, and report the disagreement rather than selecting the more flattering explainer.

C. Stability Across Training Regimes

Retraining XGBoost at $t = 100\%$ once per imbalance-handling regime and comparing global SHAP rankings pairwise (Table VI) shows Spearman correlations close to unity for all six strategy pairs, with moderate top-10 Jaccard variation: the dominant features are invariant to the training regime while the tail of the top-10 shuffles mildly. A tutor would be told substantially the same story under any of the four regimes—the explanation-side counterpart of the accuracy-side robustness in Section IV-F.

D. Drift Across Checkpoints

Explanations drift as the course progresses—and should: early checkpoints contain little behaviour, so demographic and registration priors carry relatively more weight, while later checkpoints are dominated by engagement and performance signals.

The drift is smooth and ordered: adjacent checkpoints correlate strongly, while the earliest views share only a small fraction of the full-course top-10. The explanation evolves gradually rather than jumping—evidence of a coherent underlying signal—but the early-course and end-of-course stories genuinely differ. The operational consequence, implemented in the dashboard, is that every displayed explanation is labelled with the checkpoint at which it was computed and is never transferred across checkpoints.

Taken together, the stability results answer the second research question: explanations are highly stable to training randomness, partially consistent across explainers

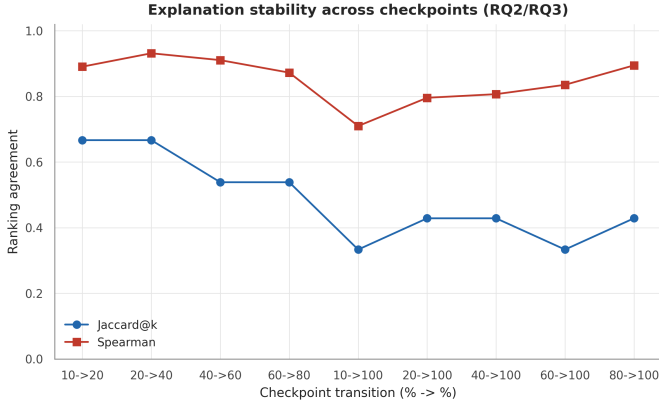


Fig. 7. Explanation drift across the six checkpoints: top-10 Jaccard overlap and Spearman rank correlation for consecutive checkpoint pairs and for each checkpoint against the full-course ($t = 100\%$) reference.

for identifiable methodological reasons, and smoothly time-varying—trustworthy, provided they are anchored on SHAP and carry their checkpoint label.

VI. Discussion and Limitations

Implications for the literature. The dual-cohort result reframes how OULAD earliness claims should be read. The full-cohort finding that reliability begins near 40% of course length is consistent with [2]; but on the population an intervention can actually reach, the same pipeline does not clear the same bar until course end, and the gap’s confidence interval excludes zero everywhere. Prior studies are not wrong—they measure outcome classification competently—but their numbers cannot be quoted as early-warning performance without naming the population, and cross-study comparisons (Table I) already founder on differing cohorts and split units. We propose dual-cohort reporting as a minimal, near-zero-cost standard: score the same predictions twice and publish both curves.

Implications for practice. The two estimands translate directly into interface and policy. The instructor dashboard operationalises the still-enrolled frame: at any checkpoint, the worklist contains only students an intervention can still reach, each with a SHAP explanation carrying its checkpoint label. Thresholds are a policy instrument selected on validation data—the default cut for a balanced workload, or the high-recall cut (0.86, test precision 0.993) when missing an at-risk student outweighs extra outreach. The threshold, like the explanation, should be stated alongside every deployment of the model.

Reproducibility. Every number in this paper is generated by a pinned, committed pipeline: the frozen test-student list and checkpoint map are version-controlled and never re-derived; 21 automated tests—including the regression test pinning the banked-assessment fix—must pass before any modelling step runs; and the paper itself

is rendered from the result tables by template substitution, so no reported value is hand-typed. The repository contains the full pipeline, the dashboard, and the scripts generating every figure and table.

Limitations. Six are stated openly. (1) Single dataset: all findings rest on one anonymised UK distance-learning dataset; transfer to other institutions, delivery modes, or platforms is untested. (2) Score availability: performance features credit a score on its submission date, but in deployment marks appear only after grading, so checkpoint-time score features are optimistic about what is knowable on the checkpoint day. (3) Source-data duplicates: 7.4% of clickstream rows are exact, key-less duplicates that the pipeline deliberately retains, so click totals may be over-counted for affected student-days. (4) LIME sample size: the SHAP–LIME agreement is estimated on $n = 100$ test students (a concession to LIME’s per-instance cost), and LIME’s treatment of one-hot categories as continuous adds noise; the estimate is coarse. (5) Borderline criterion at $t = 40\%$: even in the flattering full-cohort frame, the recall interval 0.811 (95% CI [0.798, 0.824]) touches the 0.80 criterion, so the popular “reliable from 40%” claim is weaker under honest uncertainty than point estimates suggest. (6) Evaluation-side estimand: the still-enrolled analysis re-scores a model trained on the full roster rather than retraining per checkpoint on the still-enrolled population.

Future work. Three extensions follow directly from the limitations. First, per-checkpoint censoring should be promoted from a sensitivity analysis to the primary design: training and evaluating at each checkpoint only on students still enrolled at that cutoff would make early warning the native objective of the model rather than an evaluation-side reading of it. Second, the composite at-risk label should be decomposed: Fail and Withdrawn have different behavioural signatures and call for different interventions, suggesting a multi-class or competing-risks formulation with withdrawal timing as an outcome in its own right. Third, external validity needs replication on datasets beyond OULAD, grading-lag-aware score features that respect deployment-time information availability, and larger LIME samples with additional explainers to tighten the cross-explainer agreement estimates.

VII. Conclusion

We re-examined early at-risk prediction on OULAD with a question the literature has not asked of itself: whom is the model really predicting? Under a leakage-safe, time-aware, group-aware protocol with bootstrap uncertainty, the answer is consequential. Gradient-boosted trees win a verified five-model benchmark, and on the conventional full-enrolment cohort their predictions approach reliability from 40% of course length—but a confidence-interval-honest reading makes even that claim borderline, and on the students an intervention can still reach, the reliability criterion is met only at course end, with the

between-cohort gap excluding zero at every checkpoint. Explanations, evaluated against an explicit Monte-Carlo null rather than by visual inspection, are seed-stable, partially explainer-consistent for identifiable methodological reasons, smoothly time-varying, and invariant to imbalance handling; they also reveal the mechanism of the inflation itself, since the dominant feature—current VLE silence—is precisely the channel through which already-gone students are recognised. The methodological message costs nothing to adopt and changes how results read: every early-warning claim should name its cohort, carry its uncertainty, and anchor its explanation to the checkpoint it was computed at. Under those disciplines, the pipeline presented here is accurate, interpretable, and reproducible enough to support a real early-intervention workflow.

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